# SIDDHANT PRADHAN

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## SUMMARY

Computer Engineering senior with internship and project experience building and analyzing complex hardware and software systems. Proven collaborator with strong communication skills developed through research and campus-facing roles.

## EDUCATION

**University of Massachusetts Amherst** Anticipated Spring 2027
Candidate for Bachelor of Science, Computer Engineering, **Minor:** Engineering Management, **GPA:** 3.67
**Awards**: Chancellor's Award
**Relevant coursework:** Embedded Systems, Applied ML for IoT, Systems Programming, Security Engineering, Hardware Organization & Digital Design, Information & Project Management, VLSI Design,

## PROJECTS

**CNN Digit Recognition Deployment on DE1-SoC** Spring 2026

- Deployed a convolutional neural network for handwritten digit recognition on the ARM processor of the DE1-SoC board, connecting Python-based model training and weight export to embedded real-time inference on resource-constrained hardware
- Implemented im2col and GEMM convolution to reformulate convolution as matrix multiplication, compiled and ran the inference project on the DE1-SoC, and displayed predicted digits on the seven-segment display

**Real-Time Waste Classification App** Spring 2026

- Built a smartphone-accessible app that classifies a waste item as recyclable, compostable, or landfill from a single camera photo, deploying the model through a Gradio web interface with TensorFlow Lite conversion for on-device inference
- Trained a CNN using transfer learning on MobileNetV2 in TensorFlow, combining the TrashNet and RealWaste datasets across 36,774 images and applying class weights to correct severe category imbalance

**Secure Communication System using RSA and DES** Fall 2025

- Implemented a secure end-to-end communication system on Raspberry Pi 5 combining asymmetric and symmetric cryptography for message and image transfer
- Coordinated RSA/DES workflows with socket communication and GPIO hardware feedback to ensure reliable key exchange and encrypted data transfer

**Open-source Retro PC | ECE 287 Design Project** Fall 2024

- Designed and prototyped a Raspberry Pi-based embedded system involving circuit development, PCB design, soldering, and microcontroller-based sensor integration for an open-source Retro PC project
- Evaluated design tradeoffs and modified system architecture to modernize legacy hardware demonstrating strong hardware–software integration and problem-solving skills

## WORK EXPERIENCE

**Baxter International Inc., Round Lake, IL** Summer 2026
*R&D - Hardware and Cyber Engineering Intern*

- Built an ESP32-based two-factor access control prototype for an infusion pump lockbox, replacing mechanical key access with NFC card plus PIN authentication backed by AES-256 encrypted credential storage and timestamped event logging

**Human Performance Laboratory, University of Massachusetts Amherst** Spring 2026 – Present
*Undergraduate Research Assistant*

- Developing virtual reality experiments to study truck drivers' responses to automated vehicle technology by integrating VR hardware with experimental software, troubleshooting technical issues, and supporting data collection

**Minds With Purpose Foundation, Boston, MA** Spring 2026
*AI Software Engineer Fellow*

- Built user-facing features and scalable UI architecture for a social accountability platform using modern web frameworks, resulting in streamlined navigation and measurable increases in community engagement through strategic UX improvements

## LEADERSHIP

**Executive Board Member of the ECE Student Advisory Council (ESAC)** Spring 2025 – Present

- Serving as a liaison between ECE students and faculty, addressing concerns and refining courses based on student feedback
- Assisting in creating a comprehensive semester-based ECE course guide to support students with course and elective selection

## SKILLS AND INTERESTS

**Skills:** Python, C, MATLAB, MySQL, Unix/Linux Systems Programming, ESP32, Arduino, Raspberry Pi, Altera DE1-SoC, KiCad, LTspice, Soldering, 3D Modeling, Measurement Equipment, Data Structures & Algorithms, Applied Cryptography, MS Excel
**Interests:** Guitar, Cricket, Soccer, Public Speaking, LEGO building